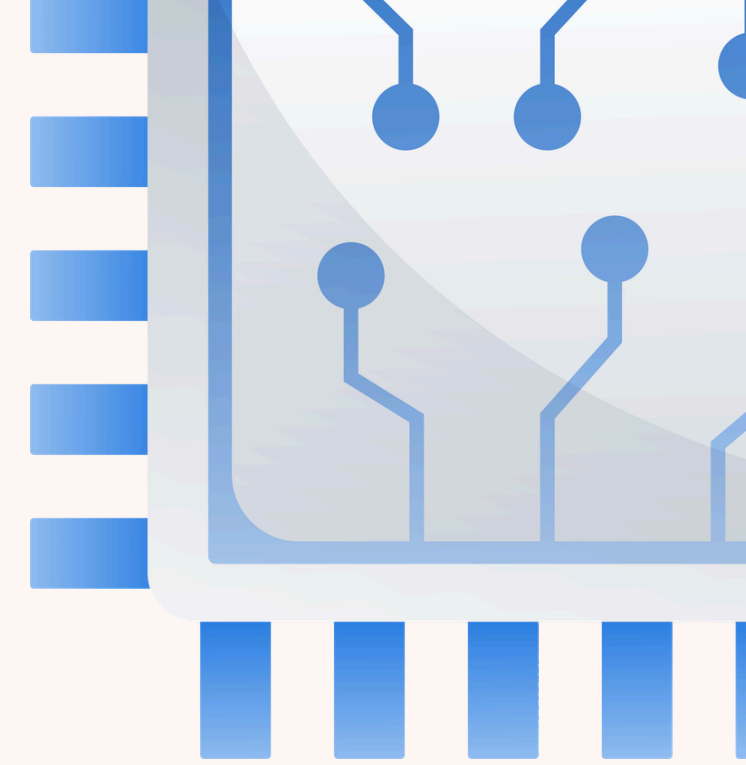
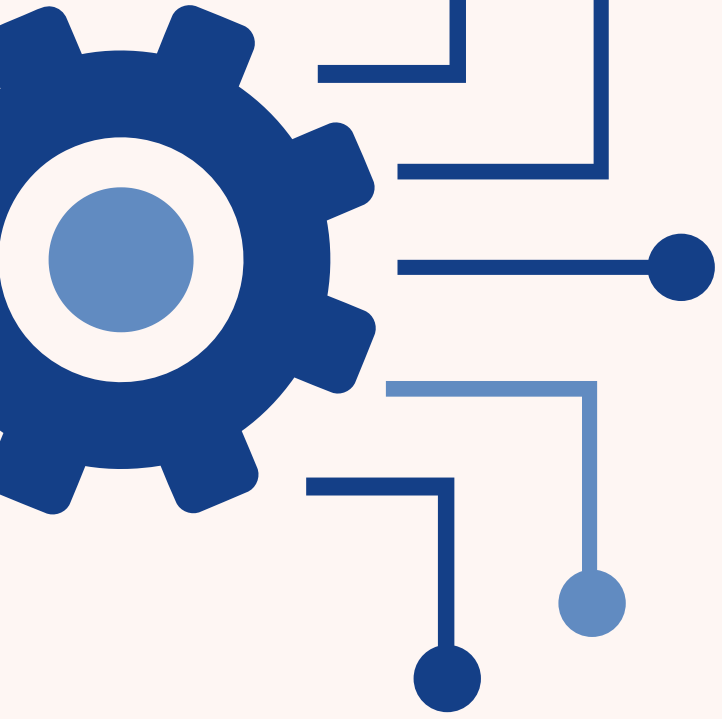


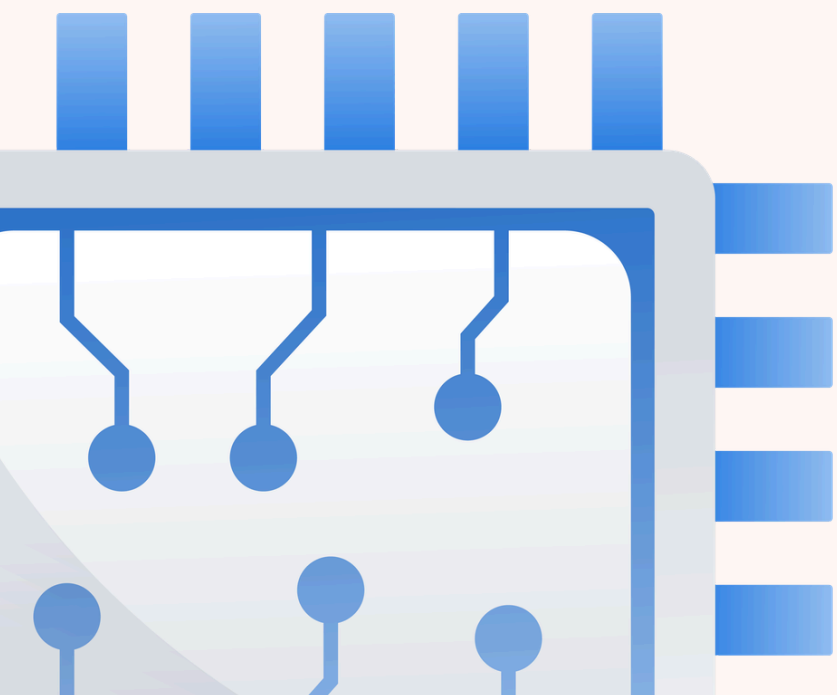


HOTEL BOOKING DEMAND

DATA ANALYSIS, STATISTICAL INFERENCE AND MACHINE LEARNING

Natalia Balicka, Kęstutis Juodis

Winter Data Science



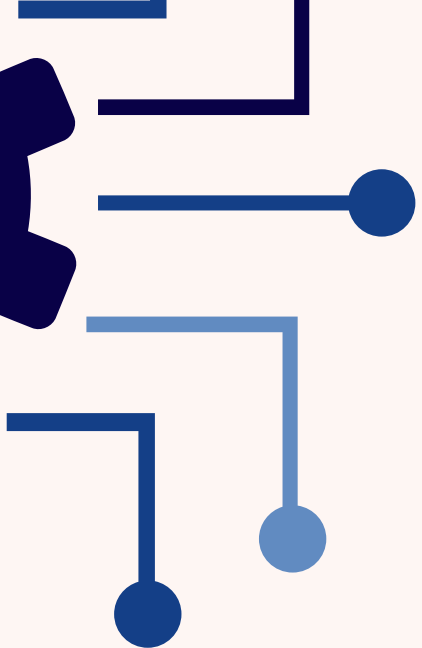
INTRODUCTION

Our group has decided to use a Hotel booking demand dataset. It has slightly over 119000 reservations, 32 initial variables , 2 hotel types - City and Resort, and targeted variable - is_canceled. 37% of all reservations were cancelled.



PROJECT OVERVIEW

PREPROCESSING



Company was removed, because 94% of values were missing. However, 32001 rows were not removed because we did not have a unique reservation identifier.

When it comes to the outliers, the lead_time maximum was 737 days, stays_in_week_nights maximum was 50, adr maximum was 5400 and ADR IQR method analyzed potential 3864 outliers. Also ADR (average daily rate) value was removed because it was negative.



PROJECT OVERVIEW

EDA

City hotel cancellation rate was 41.8% while Resort Hotel was 28%. On average reservations were made 145 days before coming, those who didn't cancel - 80 days. When it comes to the market segment groups has a cancellation rate of 61%, while direct only 15%.

Among the numerical variables, lead time had the strongest correlation with cancellation (+0.293), while special requests had a negative correlation (-0.235).

HYPOTHESIS TESTING AND STATISTICAL INFERENCE

Research Question	Statistical Test	Result	Conclusion
Does the cancellation rate differ from 35%?	One-sample proportion z-test	37.07% vs 35% $z = 14.8310$ $p < 0.001$	The null hypothesis is rejected. Cancellation rate is higher than 35%.
Does the mean booking lead time differ between cancelled and non-cancelled bookings?	Welch's t-test	144.90 vs 80.08 days $\Delta = 64.82$ days $t = 98.96$ $p < 0.001$	The null hypothesis is rejected. Cancelled bookings have a higher mean lead time.
Is hotel type associated with booking cancellation status?	Chi-square test of independence	City Hotel: 41.78% vs Resort Hotel: 27.77% $\chi^2 = 2238.34$ $p < 0.001$ V = 0.137	The null hypothesis is rejected. hotel type is associated with cancellation.

Significance level: $\alpha = 0.05$

Decision rule: if $p < 0.05 \rightarrow$ reject H_0

Additional checks : Shapiro–Wilk, Levene's test, expected frequencies, Cramér's V.

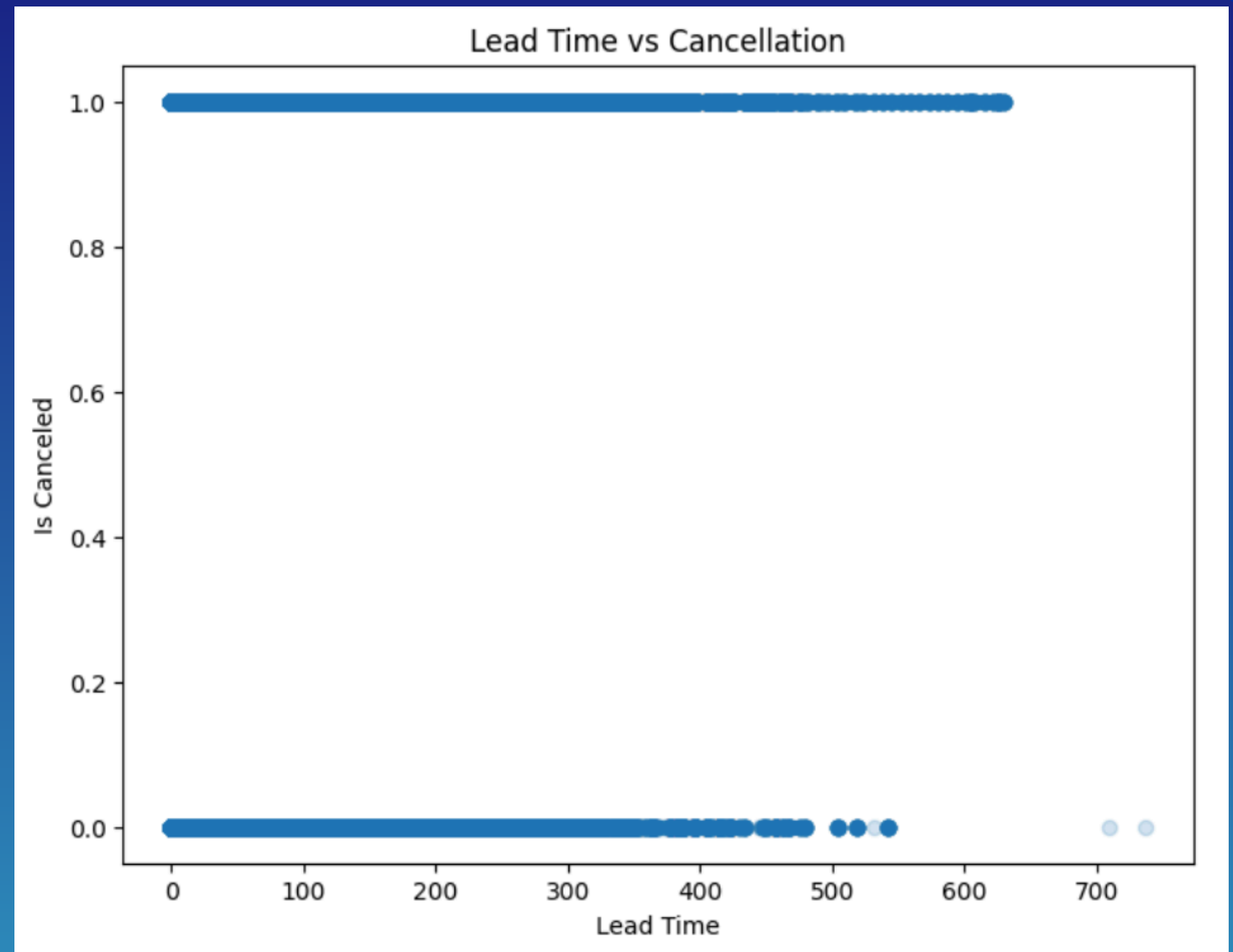
LINEAR AND BINARY LOGISTIC REGRESSION

Linear Regression – Predicting ADR

- Target: Average Daily Rate (ADR)
- $R^2 = 0.187$
- Model explains 18.7% of ADR variation
- Limited explanatory power

Logistic Regression – Predicting Cancellation

- Target: Booking cancellation (is_canceled)
- Accuracy: 81.4%
- ROC-AUC: 0.889
- Recall: 64.8%
- Strong overall discrimination, but some cancellations are missed



CLASSIFICATION WITH DECISION TREES

	Logistic Regression	Decision Tree
Accuracy	81.4%	80.8%
Recall	64.8%	69.0%
F1	72.0%	72.8%
ROC-AUC	88.9%	83.9%

Conclusion: When comparing logistic regression and decision tree results, the decision tree achieves higher overall recall and F1-score, while logistic regressions achieves higher accuracy and ROC-AUC scores. The best performing tree used a depth of 20.

K-MEANS CLUSTERING: METHOD AND SELECTION OF K

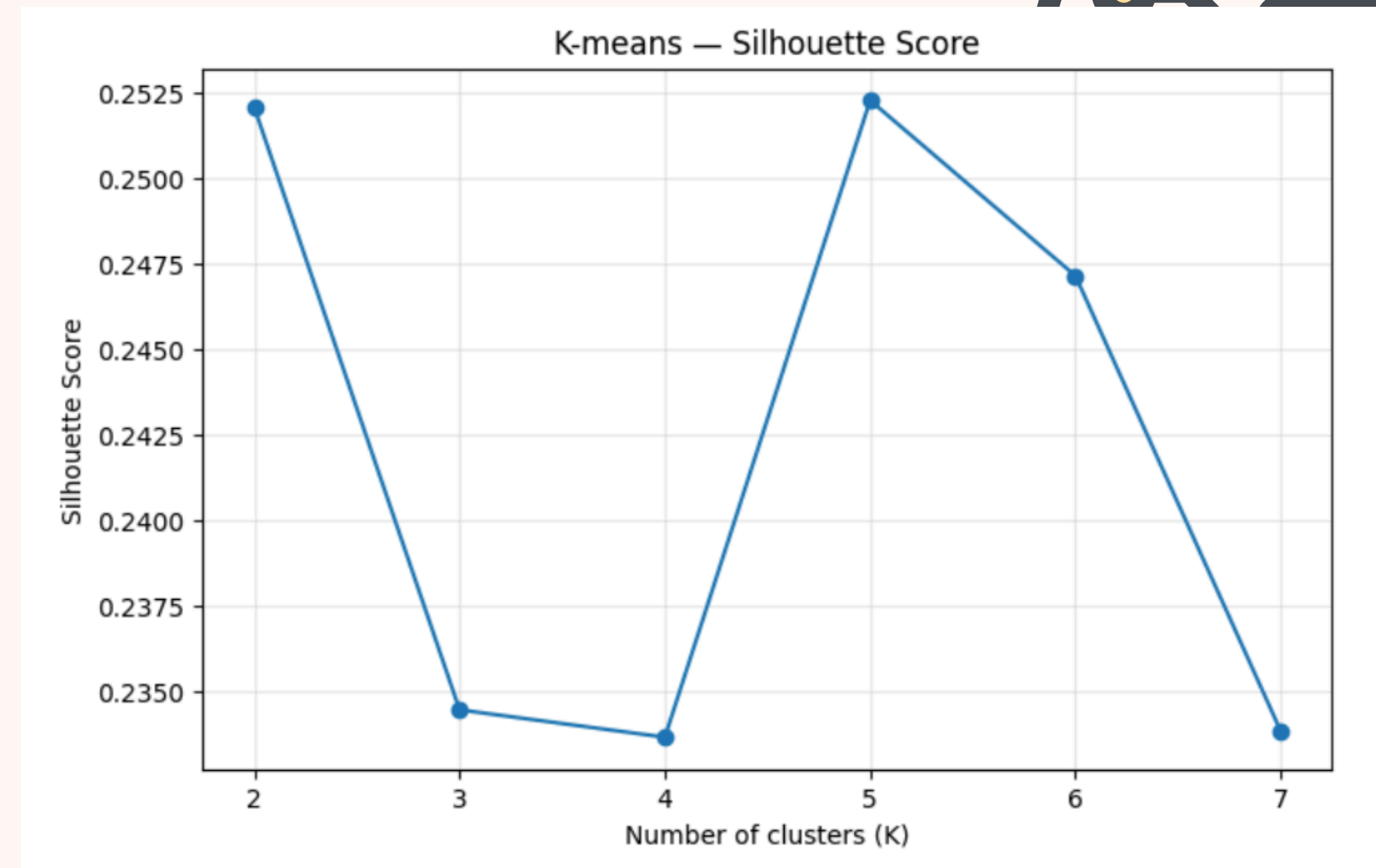
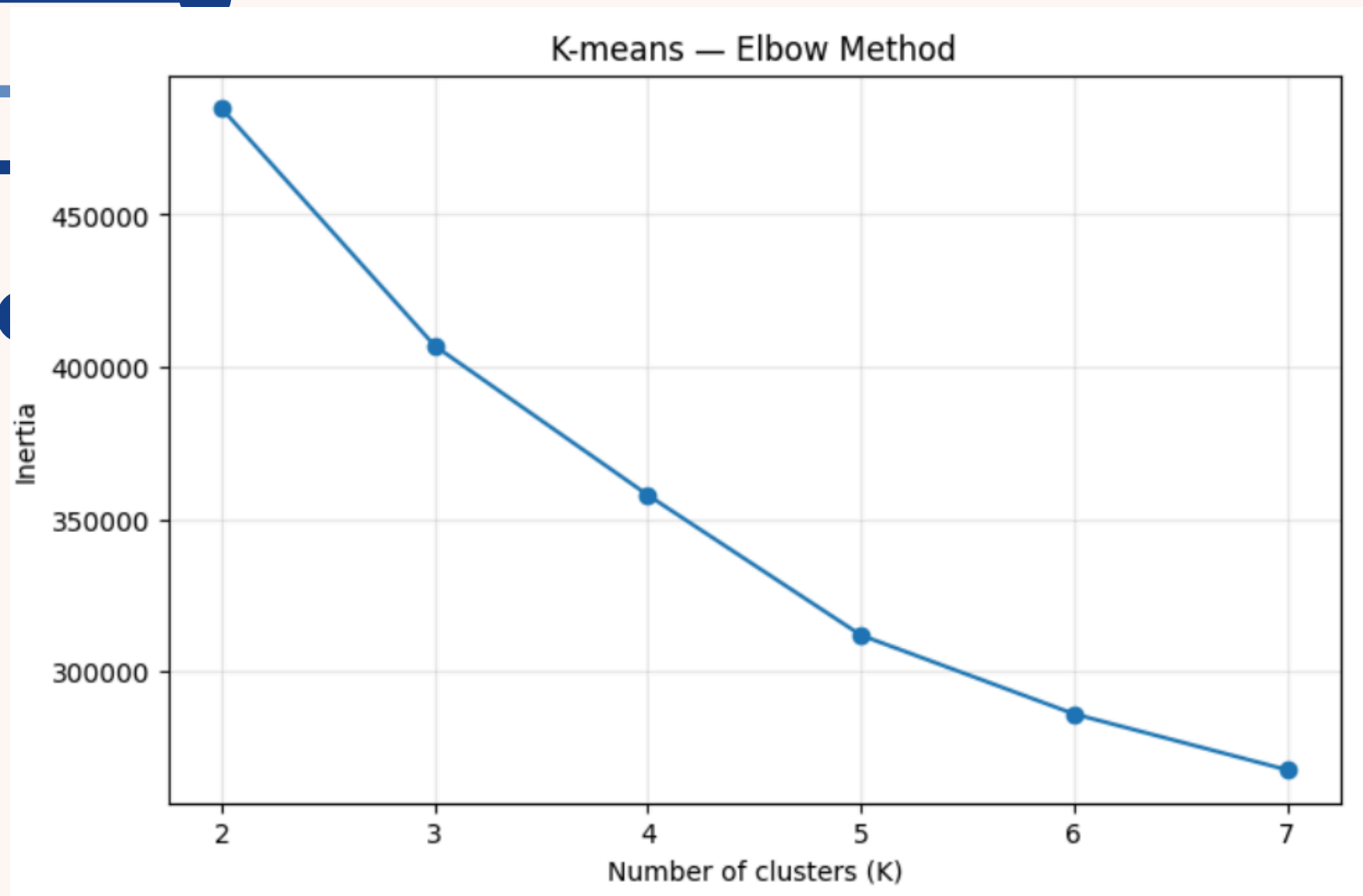
RESEARCH QUESTION : Can hotel bookings be meaningfully grouped into distinct booking profiles based on their characteristics?

FEATURES : lead time, ADR, total nights, total guests, special requests

OBSERVATIONS: 119,205

METHODS: K-means on standardized data

TESTED: $K = 2-7$



Selected solution: $K = 5$

Silhouette Score = 0.2523

$K = 5$ was selected based on the highest Silhouette Score and was also supported by the Elbow curve.

FIVE HOTEL BOOKING PROFILES

Number of bookings Percentage (%)

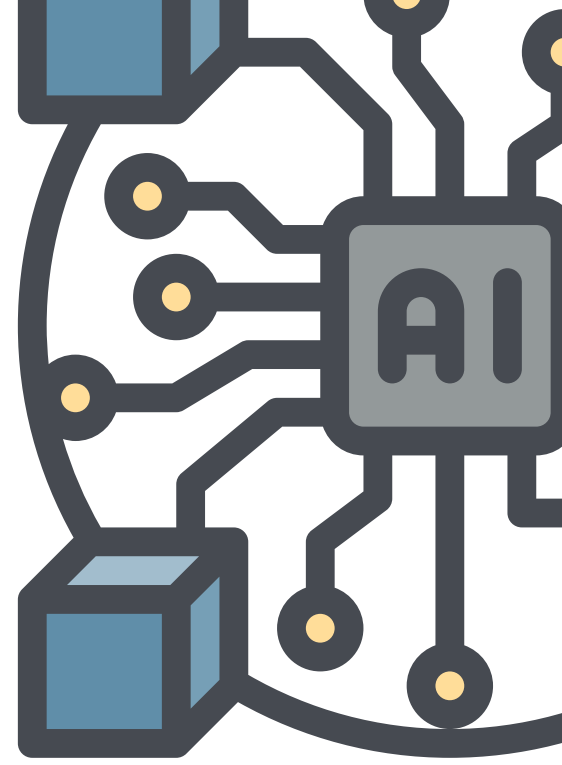
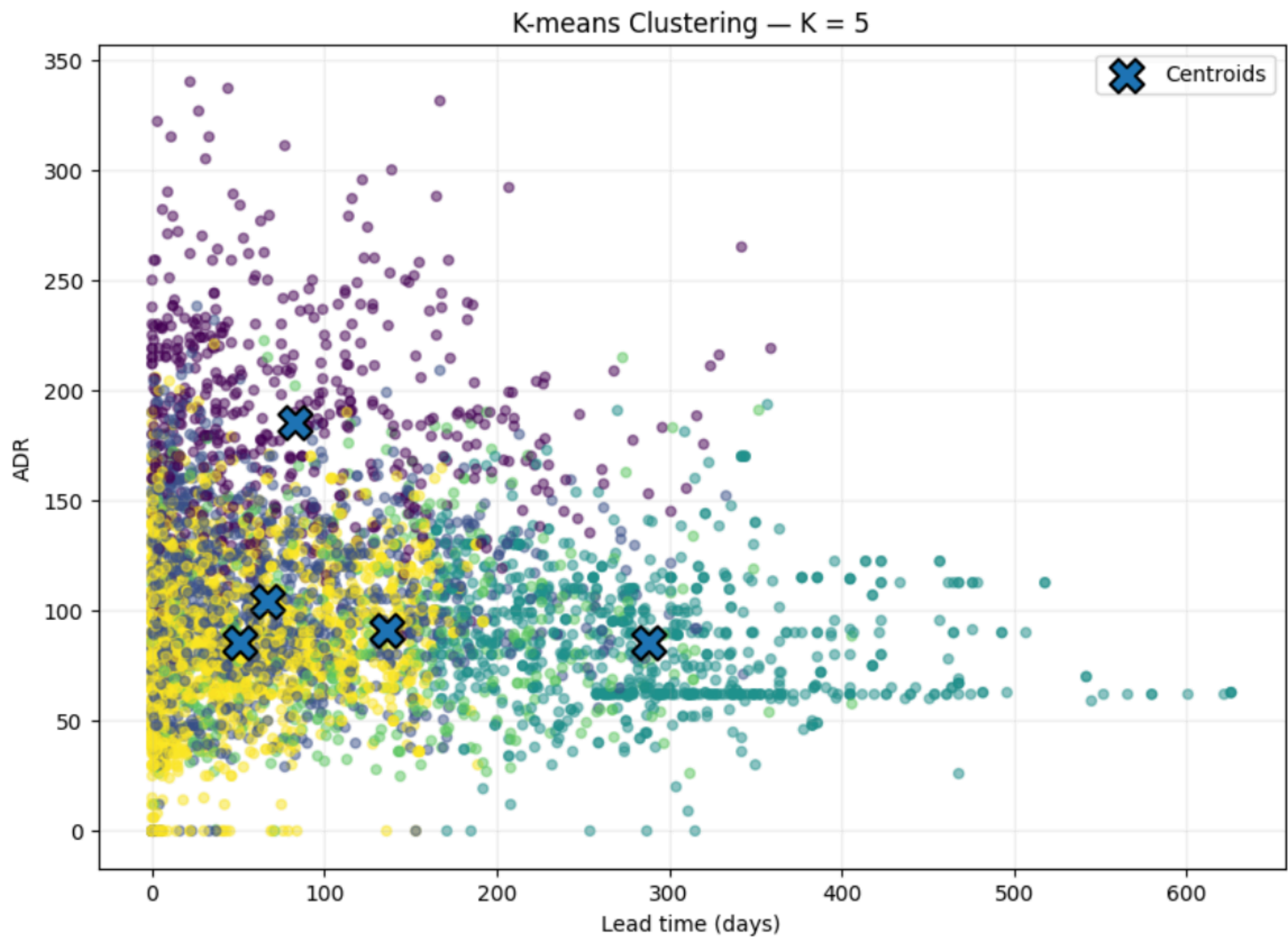


cluster

0	13217	11.09
1	30759	25.80
2	18893	15.85
3	11654	9.78
4	44682	37.48

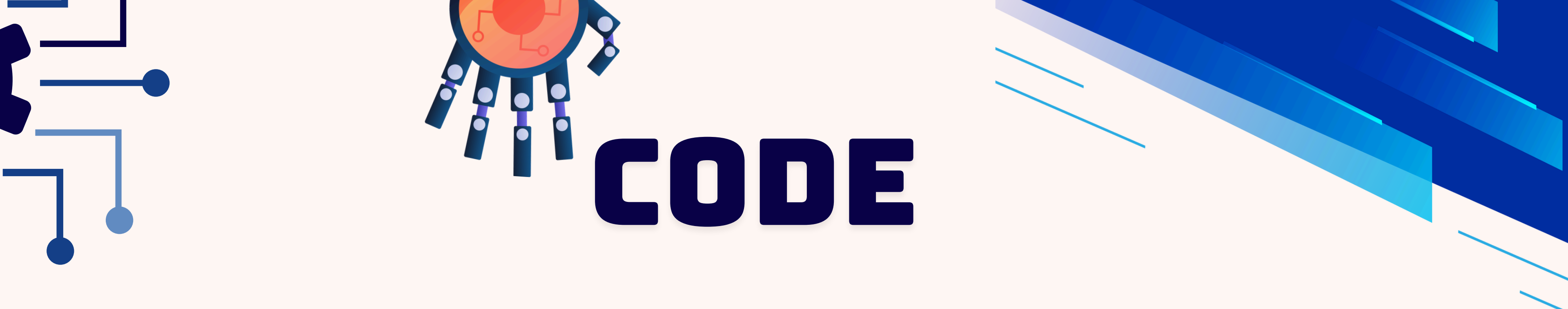
- **Cluster 0:** ADR 185.70, 3.10 guests → clearly higher-priced bookings and larger groups.
- **Cluster 1:** 1.51 special requests → a cluster characterized by a relatively high number of special requests.
- **Cluster 2:** lead_time 287.60 days → bookings made very far in advance.
- **Cluster 3:** 8.74 nights → clearly longer stays.
- **Cluster 4:** 50.52 days, 2.42 nights, 1.63 guests, 0.07 requests → shorter and simpler bookings, made relatively close to the arrival date.

K-means Clustering - K = 5



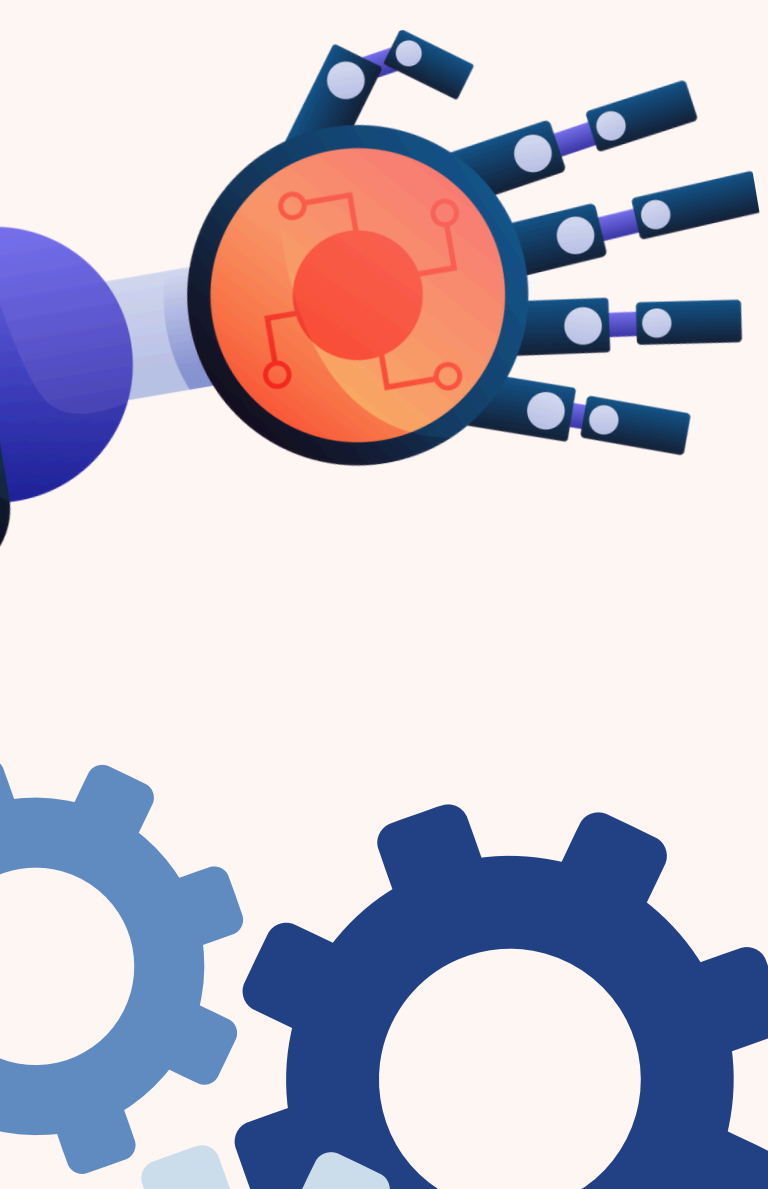
BOOKING PROFILES AND CANCELLATION BEHAVIOUR

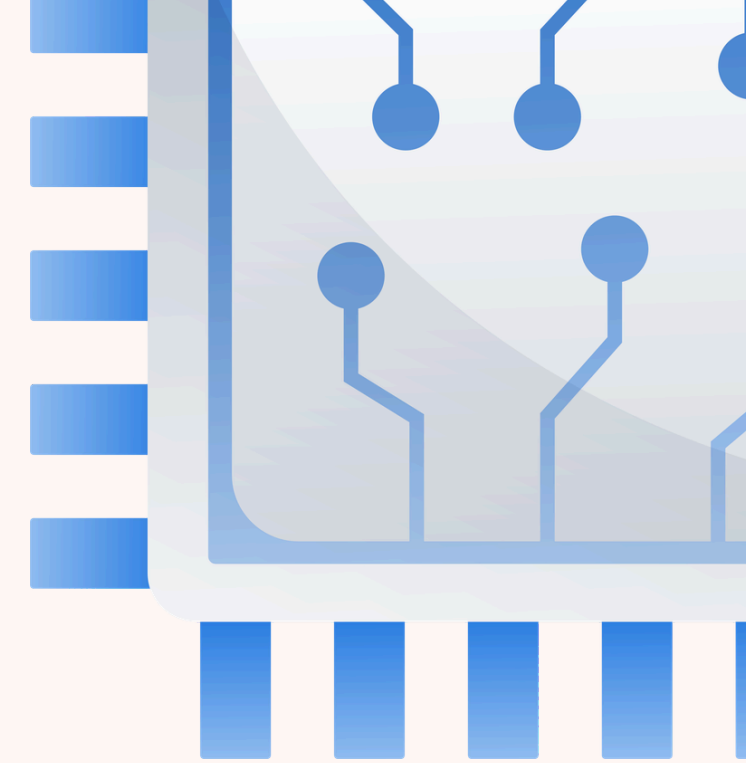
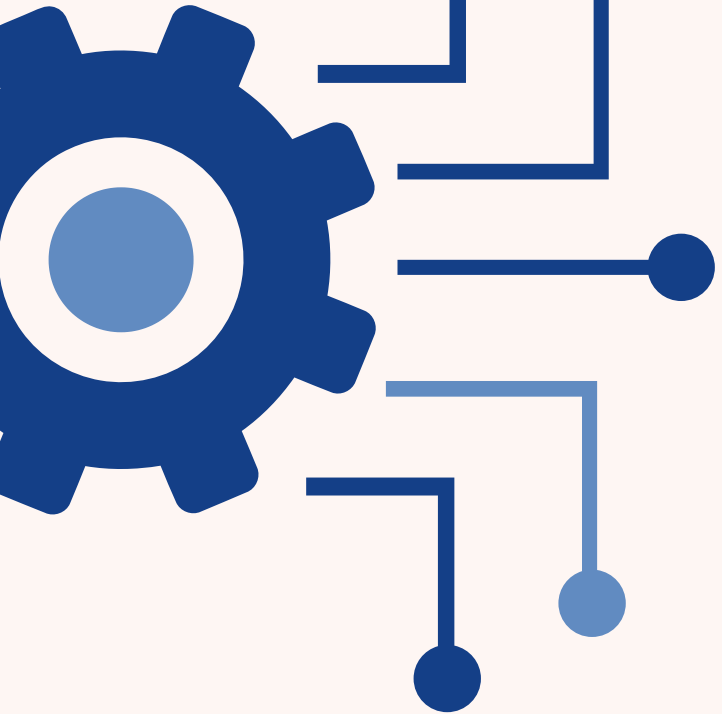
- Cluster 2 has the highest cancellation rate (63.85%).
- It represents bookings made very far in advance.
- Cluster 1 has the lowest cancellation rate (20.23%).
- It is characterized by a relatively high number of special requests.
- Cancellation status was not used to create the clusters — it was analysed afterwards to describe the identified booking profiles.



CODE

<https://github.com/balickanatalka/Hotel-Booking-Demand---Analysis>





THANK YOU

Natalia Balicka, Kęstutis Juodis

